

Assignment 3: Swiss Cheese Test Suite

Smart Water Capstone

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1 Objective

Use AI-generated hydrology code and prove layered verification catches at least one AI hallucination or false assumption.

2 Swiss Cheese layers

1. Human / AI review (units, formulas)
2. Unit & boundary pytest (33 capstone tests)
3. Physical validity (`src/validation.py`)
4. Integration tests
5. Coverage gate (>80% on `src/`)

3 Boundary test examples

Test	Category	Rule
<code>test_runoff_zero_rainfall</code>	Boundary	$P = 0 \Rightarrow Q = 0$
<code>test_runoff_below_initial_abstraction</code>	Boundary	$P < I_a \Rightarrow Q = 0$
<code>test_runoff_not_exceed_rainfall_high_cn</code>	Physical	$Q \leq P$ for CN 60/80/95
<code>test_runoff_negative_p_raises</code>	Invalid	$P < 0$ raises error
<code>test_validate_runoff_fail_case</code>	Physical	$Q > P$ fails validator

4 Results

33 passed; 96% coverage on `src/`. Evidence: `lab_reports/week8_session_a_terminal.png` (Session 8A snapshot; expanded to 33 tests for Assignment 3 regression cases).

5 Hallucination (summary)

AI assumption: “For CN=95, $P = 5$ mm always gives zero runoff.”

Caught by: Ia formula review + pytest edge design.

Fix: Correct zero-runoff test ($P = 2$ mm, CN=80); add $Q \leq P$ sweep.

Full case study: `assignment3/HALLUCINATION_CASE_STUDY.md`.

6 Repository

<https://github.com/mahmud456alhasan-debug/smart-water-capstone>